

BALARAM SARKAR

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Education

National Institute of Technology, Rourkela

Bachelor of Technology in Computer Science and Engineering (CGPA - 7.37)

August 2023 – Present

Rourkela, Odisha

Gangarampur High School, Gangarampur

WBCHSE, Science (PCMB) (Percentage - 87.6)

June 2022

Gangarampur, West Bengal

Kadihat Belbari High School, Gangarampur

WBBSE (Percentage - 95.42)

July 2020

Gangarampur, West Bengal

Relevant Coursework

- Data Structures
- Algorithms
- Operating Systems
- Database Management System
- Object Oriented Programming
- Computer Networks

Projects

Multi-Agent Research Pipeline

May 2026 | [🌐 GitHub](#) | [Live Demo](#)

Python, LangChain, LangGraph, OpenAI GPT-4o, Tavily, Streamlit

- **Architected a multi-agent AI pipeline** with four specialized agents (Search, Reader, Writer, Critic) using **LangChain** and **LangGraph's ReAct framework** for autonomous end-to-end research generation.
- **Integrated Tavily web search** and a custom **BeautifulSoup scraper** as LangChain tools, enabling real-time information retrieval and deep content extraction from live web sources.
- **Designed a writer-critic feedback loop** using structured prompt chains, producing scored research reports with quantitative evaluation and actionable improvement feedback.

Paste Vault | React, Redux Toolkit, JavaScript, LocalStorage

Dec 2025 | [🌐 GitHub](#) | [Live Demo](#)

- **Engineered a React-based web application** utilizing **functional components** and **hooks** for efficient user-generated content management.
- **Architected global state management** via **Redux Toolkit** with **persistent storage** through **localStorage** to ensure seamless data retention across sessions.
- **Optimized code maintainability and scalability** by implementing **modular components** and a **clean project architecture**, reducing component coupling.

Technical Skills

Languages: Python, C++, JavaScript, HTML, CSS, SQL

AI / ML: LangChain, LangGraph, OpenAI API, Tavily

Libraries & Frameworks: React, Redux Toolkit, Node.js, Express.js, Tailwind CSS, Streamlit, BeautifulSoup

Tools & Platforms: Git, GitHub, Vite, VS Code, Vercel, Streamlit Cloud

Leadership / Extracurricular

NSS Volunteer – NITR

August 2023 – December 2023

Coordinator

NIT, Rourkela

- Spearheaded **community engagement initiatives** under **Swachh Bharat Abhiyan**, managing **200+ volunteers** and large-scale campus clean-up drives.

Innovision

November 2024

Coordinator

NIT, Rourkela

- Coordinated **event operations, team management, and cross-functional collaboration** for Innovision, one of the **largest annual techno-management fests in Eastern India**, showcasing technical competitions, workshops, and exhibitions.